



CODECHEF

Here's your monthly dose of coding puzzles from India's biggest online coding contest, now in print!

Solve this month's puzzle, and you stand the chance of being one of the three lucky people who could win a cash prize of Rs 1,00,000 each!

The September edition puzzle:

There is a 50 cm long line of ants moving ahead. The last ant in the line had an urgent message to give to the queen ant, which is moving at first position. So while the line is moving, the last ant runs ahead, reaches the first ant (the queen) and passes on the message. And without stopping, it runs back to its original position. (Assume that no time is lost in giving the message.) During this time, the whole line has moved ahead by 50 cm. The question is: how much distance did the last ant cover in that time? Assume that it ran the whole distance at uniform speed.

The Solution:

Let x be the distance travelled by the line when the last ant reaches the queen ant.

Let u be the speed of the ant line and v be the speed of the last ant.

Let t_1 be the time taken by the ant to reach the queen ant.

Let t_2 be the time taken by the ant to return to its original position in the line, after delivering the message to the queen ant.

then,

$$50 + x = v * t_1 \quad (\text{for the last ant}) \dots (1)$$

$$x = u * t_1 \quad (\text{for the ant line}) \dots (2)$$

Also, during the return journey,

$$50 - x = u * t_2 \quad (\text{for the ant line}) \dots (3)$$

$$x = v * t_2 \quad (\text{for the last ant}) \dots (4)$$

Then, from Equations 1 and 2, you get,

$$(50 + x) / x = v/u \dots (5)$$

Also, from Equations 3 and 4, you get,

$$x / (50 - x) = v/u \dots (6)$$

On comparing Equations 5 and 6, you get,

$$x^2 = 50^2 - x^2 \Rightarrow x = \sqrt{1250} = 35.36$$

$$x = 35.36$$

The total distance travelled by the last ant is $50 + 2x$.

$$50 + 2(35.36) = 120.72.$$

And the winners this month are:

- Shakti Rath
- Mitanshu Bakshi
- Kshitij Rastogi

Here's your CodeChef 'Puzzle of the Month':

Here are the conditions of the magic number:

1. If the magic number was a multiple of 2, then it was a number from 50 through 59.
2. If it was not a multiple of 3, then it was a number from 60 through 69.
3. If the magic number was not a multiple of 4, then it was a number from 70 through 79.

What was the magic number?.

Looking for a little extra cash?

Solve this month's CodeChef Challenge puzzle and send in your answer to codechef@efyindia.com or lly@codechef.com by October 15, 2012. Three lucky winners get to win some awesome prizes both in cash and other merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou>. 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the October Challenge that takes place from the 1st to the 11th, to win cash prizes of up to Rs 20,000. Also keep visiting CodeChef and participate in the multiple programming contests that take place on the website throughout the month.